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I am completing my PhD in theoretical physics at EPFL, with a defense scheduled in August 2026. My work focuses on turning abstract questions into well-defined mathematical problems and producing quantitative answers, using a combination of analytical reasoning and computational methods. I am transitioning toward machine learning research, drawn in particular by questions about the mathematical structure underlying learning and generalisation, and by the prospect of developing methods that can genuinely accelerate scientific discovery.

Education

2022 – ... ◇ **PhD, EPFL - Fields and Strings Laboratory**

Supervised by João Penedones. Research on non-perturbative quantum field theory and quantum gravity, spanning the S-matrix bootstrap (bounding scattering amplitudes in 2D QFT via semidefinite programming) and the large- N matrix model approach to emergent spacetime. Combined pen-and-paper analysis with computational workflows in Python and Wolfram Language: large-scale SDP (SDPB) on HPC clusters, Hamiltonian truncation, spectral methods.

Thesis: *What is spacetime made of?*

2017 – 2022 ◇ **Bachelor and Master of Physics, EPFL**

Skills

Coding ◇ Python (NumPy, SciPy, PyTorch, JAX/Flax/Optax, matplotlib), Wolfram Language, $\text{\LaTeX}$, C++. HPC cluster workflows (SLURM, Docker), semidefinite programming (SDPB).

Languages ◇ Native in French, fluent in English, B2 in German.

General ◇ Problem solving, scientific communication, teaching, project supervision.

Research Output

◇ 5 publications in high energy theoretical physics (JHEP), see INSPIRE profile <https://inspirehep.net/authors/2612961>.

◇ Multiple invited talks at international workshops and universities

Experience in Machine Learning

◇ Built a variational neural solver for large- N Hermitian matrix models. The eigenvalue density is parametrised by an MLP trained to minimise the effective matrix model action, recovering exact analytic solutions in benchmark cases and handling expected phase transitions. Implemented in both PyTorch and JAX.

◇ Implemented a decoder-only transformer following the GPT-2 architecture (12 layers, 12 heads, 768-dim embeddings) and a character-level language model with knowledge distillation experiments, both from scratch in PyTorch.

◇ Participated at the AI for High Energy Physics EPFL workshop. Hands-on training with JAX and modern neural network architectures, with applications to physics problems.

Teaching and scientific activities

◇ **Teaching Assistant, EPFL** - 10+ courses on physics and mathematics at the bachelor and master level. Supervised two master students on their thesis. Gave lectures, designed projects and supervised students for EPFL summer schools.

◇ **Peer Reviewer, JHEP** - Served as peer reviewer for JHEP, evaluating manuscripts on quantum gravity for scientific rigor, methodology, and novelty.

◇ **Research seminar organizer, EPFL** End-to-end coordination of weekly seminar series, including speaker selection, logistics, chairing, and communication.